



7085 - Dust Evolution in Molecular Cores

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	b68 plan v7	NIRSpec MultiObject Spectroscopy	(1) B68-MASTER
	2	b335 plan v1	NIRSpec MultiObject Spectroscopy	(2) B335-MASTER
	3	l694 v2	NIRSpec MultiObject Spectroscopy	(3) SELECTION-L694

ABSTRACT

We propose to constrain the initial stages of grain growth in early sites of star formation. This objective will be achieved by obtaining NIRSpec MSA spectroscopic observations of background stars behind three dense molecular cores of different evolutionary stages. By employing background stars, we will avoid uncertainties arising when embedded sources, with uncertain IR excesses, are used. The proposed observations will provide unprecedented spectroscopic information on the systematic variation of the extinction curve with depth into the cloud, at levels of extinction that were not possible to probe before JWST.

OBSERVING DESCRIPTION

We propose to perform NIRSpec Multi-Shutter Array (MSA) spectroscopic observations of background stars as seen behind three Galactic cores (B68, L694-2, B335) at different evolutionary stages. Our catalogs of background stars contain large numbers of stars through which we can study the three cores (427, 387 and 331 stars behind the B68, L694-2 and B335 cores, respectively). The observations will be executed in the G140H/F100LP, G235H/F170LP and the G395H/F290LP filters in the high-spectral resolution mode.

Given the astrometric accuracy of our source catalogs, we selected the MSATA Target Acquisition method. In addition, the aperture PA is set to 50.0 (B68) and 0.0 (B335/L694-2). We will use 13 groups per integration and 1 integration per exposure, with a NRSIRS2 readout pattern, which allows optimizing data volume. In order to obtain a uniform spatial and spectral coverage for the targets in each of the molecular cores, as well as to mitigate leakage signal through the MSA, we will use two configurations per core with the recommended 3-shutter slitlet nod pattern.

Finally, using the imaging mirror of NIRSpec, we will take confirmation images, in the Target Acquisition and New MSA Config mode, with 10 groups per integration, 1 integration per exposure, and by using the NRSIRS2RAPID readout pattern. These confirmation images will be used to analyze the position of science sources through the open shutter slitlets.

The total requested time is 24.10 hrs, with 14.48 hrs of science time, for a corresponding efficiency of 60%.

Proposal 7085 - Targets - Dust Evolution in Molecular Cores

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	B68-MASTER	RA: 17 22 39.1581 (260.6631587d) Dec: -23 48 22.12 (-23.80614d) Equinox: J2000		
	Comments: Description=[]				
	(2)	B335-MASTER	RA: 19 36 58.9344 (294.2455600d) Dec: +07 34 20.17 (7.57227d) Equinox: J2000		
	Comments: Description=[]				
	(3)	SELECTION-L694	RA: 19 41 5.7374 (295.2739058d) Dec: +10 58 59.39 (10.98316d) Equinox: J2000		
	Comments: Description=[]				

Proposal 7085 - Observation 1 - Dust Evolution in Molecular Cores

Observation	Proposal 7085, Observation 1: b68 plan v7											Tue Mar 11 20:05:26 GMT 2025
	Diagnostic Status: Warning											
Observing Template: NIRSpec MultiObject Spectroscopy												
Diagnostics	(b68 plan v7 (Obs 1)) Warning (Form): Config c1 (#1) has 1 filler slits affected by failed closed shutters.											
	(b68 plan v7 (Obs 1)) Warning (Form): Config c1 (#1) has 2 primary slit traces affected by failed open shutters.											
	(b68 plan v7 (Obs 1)) Warning (Form): Config c1 (#2) has 1 filler slits affected by failed closed shutters.											
	(b68 plan v7 (Obs 1)) Warning (Form): Config c1 (#2) has 2 primary slit traces affected by failed open shutters.											
	(b68 plan v7 (Obs 1)) Warning (Form): Config c1 (#3) has 1 filler slits affected by failed closed shutters.											
	(b68 plan v7 (Obs 1)) Warning (Form): Config c1 (#3) has 2 primary slit traces affected by failed open shutters.											
	(b68 plan v7 (Obs 1)) Warning (Form): Config c2 (#4) has 1 filler slit traces affected by failed open shutters.											
	(b68 plan v7 (Obs 1)) Warning (Form): Config c2 (#4) has 3 primary slit traces affected by failed open shutters.											
	(b68 plan v7 (Obs 1)) Warning (Form): Config c2 (#5) has 1 filler slit traces affected by failed open shutters.											
	(b68 plan v7 (Obs 1)) Warning (Form): Config c2 (#5) has 3 primary slit traces affected by failed open shutters.											
	(b68 plan v7 (Obs 1)) Warning (Form): Config c2 (#6) has 1 filler slit traces affected by failed open shutters.											
	(b68 plan v7 (Obs 1)) Warning (Form): Config c2 (#6) has 3 primary slit traces affected by failed open shutters.											
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections		Miscellaneous			
	(1)	B68-MASTER	RA: 17 22 39.1581 (260.6631587d) Dec: -23 48 22.12 (-23.80614d) Equinox: J2000									
Comments: Description=[]												
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List		Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold
	MSATA	false		After Target ACQ and New MSA Config	MSA Center	high-extinction (427 sources)		B68-MASTER (2603 sources)		jwst-nirspec-hr		1.5
Reference Stars												

Proposal 7085 - Observation 1 - Dust Evolution in Molecular Cores

Confirmation											
	#	Confirmation Type		Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp		Conf. Total Integrations		Conf. Total Exposure Time	
Spectral Elements	1	c1		NRSIRS2RAPID	10	1		1		160.478	
	2	c2		NRSIRS2RAPID	10	1		1		160.478	
	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140H/F100LP)	c1	3 Shutter Slitlet	260.684735375 Degrees - 23.829456666666 66 Degrees	49.991243438339 815			3	3	2888.6
	2	2 (G235H/F170LP)	c1	3 Shutter Slitlet	260.684735375 Degrees - 23.829456666666 66 Degrees	49.991243438339 815			3	3	2888.6
	3	3 (G395H/F290LP)	c1	3 Shutter Slitlet	260.684735375 Degrees - 23.829456666666 66 Degrees	49.991243438339 815			3	3	2888.6
Special Requirements	4	1 (G140H/F100LP)	c2	3 Shutter Slitlet	260.69198795833 336 Degrees - 23.834769166666 66 Degrees	49.988305280561 77			3	3	2888.6
	5	2 (G235H/F170LP)	c2	3 Shutter Slitlet	260.69198795833 336 Degrees - 23.834769166666 66 Degrees	49.988305280561 77			3	3	2888.6
	6	3 (G395H/F290LP)	c2	3 Shutter Slitlet	260.69198795833 336 Degrees - 23.834769166666 66 Degrees	49.988305280561 77			3	3	2888.6
MSA Planned Aperture PA 50.0000 to 50.0000 Degrees (V3 271.4254303 to 271.4254303)											

Proposal 7085 - Observation 2 - Dust Evolution in Molecular Cores

Observation	Proposal 7085, Observation 2: b335 plan v1											Tue Mar 11 20:05:27 GMT 2025
	Diagnostic Status: Warning											
Observing Template: NIRSpec MultiObject Spectroscopy												
Diagnostics	(b335 plan v1 (Obs 2)) Warning (Form): Config c1 (#1) has 4 primary slit traces affected by failed open shutters.											
	(b335 plan v1 (Obs 2)) Warning (Form): Config c1 (#2) has 4 primary slit traces affected by failed open shutters.											
	(b335 plan v1 (Obs 2)) Warning (Form): Config c1 (#3) has 4 primary slit traces affected by failed open shutters.											
	(b335 plan v1 (Obs 2)) Warning (Form): Config c2 (#4) has 1 filler slit traces affected by failed open shutters.											
	(b335 plan v1 (Obs 2)) Warning (Form): Config c2 (#4) has 2 primary slit traces affected by failed open shutters.											
	(b335 plan v1 (Obs 2)) Warning (Form): Config c2 (#4) has 2 primary slits affected by failed closed shutters.											
	(b335 plan v1 (Obs 2)) Warning (Form): Config c2 (#5) has 1 filler slit traces affected by failed open shutters.											
	(b335 plan v1 (Obs 2)) Warning (Form): Config c2 (#5) has 2 primary slit traces affected by failed open shutters.											
	(b335 plan v1 (Obs 2)) Warning (Form): Config c2 (#5) has 2 primary slits affected by failed closed shutters.											
	(b335 plan v1 (Obs 2)) Warning (Form): Config c2 (#6) has 1 filler slit traces affected by failed open shutters.											
	(b335 plan v1 (Obs 2)) Warning (Form): Config c2 (#6) has 2 primary slit traces affected by failed open shutters.											
	(b335 plan v1 (Obs 2)) Warning (Form): Config c2 (#6) has 2 primary slits affected by failed closed shutters.											
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(2)	B335-MASTER	RA: 19 36 58.9344 (294.2455600d) Dec: +07 34 20.17 (7.57227d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold	
	MSATA	false		After Target ACQ and New MSA Config	MSA Center	B335-high extinction (331 sources)	B335-MASTER (2448 sources)		jwst-nirspec-hr		1.5	
Reference Stars												

Proposal 7085 - Observation 2 - Dust Evolution in Molecular Cores

Confirmation											
	#	Confirmation Type		Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp		Conf. Total Integrations		Conf. Total Exposure Time	
Spectral Elements	1	c1		NRSIRS2RAPID	10	1		1		160.478	
	2	c2		NRSIRS2RAPID	10	1		1		160.478	
	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140H/F100LP)	c1	3 Shutter Slitlet	294.25719183333 337 Degrees 7.58091111111111 11 Degrees	0.0015279539441 012385			3	3	2888.6
	2	2 (G235H/F170LP)	c1	3 Shutter Slitlet	294.25719183333 337 Degrees 7.58091111111111 11 Degrees	0.0015279539441 012385			3	3	2888.6
	3	3 (G395H/F290LP)	c1	3 Shutter Slitlet	294.25719183333 337 Degrees 7.58091111111111 11 Degrees	0.0015279539441 012385			3	3	2888.6
Special Requirements	4	1 (G140H/F100LP)	c2	3 Shutter Slitlet	294.25214779166 663 Degrees 7.59091111111111 11 Degrees	8.8344051173618 3E-4			3	3	2888.6
	5	2 (G235H/F170LP)	c2	3 Shutter Slitlet	294.25214779166 663 Degrees 7.59091111111111 11 Degrees	8.8344051173618 3E-4			3	3	2888.6
	6	3 (G395H/F290LP)	c2	3 Shutter Slitlet	294.25214779166 663 Degrees 7.59091111111111 11 Degrees	8.8344051173618 3E-4			3	3	2888.6
MSA Planned Aperture PA 0.0000 to 0.0000 Degrees (V3 221.4254303 to 221.4254303)											

Proposal 7085 - Observation 3 - Dust Evolution in Molecular Cores

Observation	Proposal 7085, Observation 3: I694 v2											Tue Mar 11 20:05:27 GMT 2025		
	Diagnostic Status: Warning													
	Observing Template: NIRSpec MultiObject Spectroscopy													
Diagnostics	(I694 v2 (Obs 3)) Warning (Form): Config c1 (#1) has 1 primary slits affected by failed closed shutters.													
	(I694 v2 (Obs 3)) Warning (Form): Config c1 (#1) has 3 primary slit traces affected by failed open shutters.													
	(I694 v2 (Obs 3)) Warning (Form): Config c1 (#2) has 1 primary slits affected by failed closed shutters.													
	(I694 v2 (Obs 3)) Warning (Form): Config c1 (#2) has 3 primary slit traces affected by failed open shutters.													
	(I694 v2 (Obs 3)) Warning (Form): Config c1 (#3) has 1 primary slits affected by failed closed shutters.													
	(I694 v2 (Obs 3)) Warning (Form): Config c1 (#3) has 3 primary slit traces affected by failed open shutters.													
	(I694 v2 (Obs 3)) Warning (Form): Config c2 (#4) has 1 primary slits affected by failed closed shutters.													
	(I694 v2 (Obs 3)) Warning (Form): Config c2 (#5) has 1 primary slits affected by failed closed shutters.													
	(I694 v2 (Obs 3)) Warning (Form): Config c2 (#6) has 1 primary slits affected by failed closed shutters.													
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous					
	(3)	SELECTION-L694	RA: 19 41 5.7374 (295.2739058d) Dec: +10 58 59.39 (10.98316d) Equinox: J2000											
	Comments: Description=[]													
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID			
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788				
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List		Filler Candidate List		Spectral Overlap Map	Spectral Overlap Threshold			
	MSATA	false		After Target ACQ and New MSA Config	MSA Center	SELECTION-L694 (387 sources)		SELECTION-L694 (387 sources)		jwst-nirspec-hr	1.5			
Reference Stars														
Confirmation	#	Confirmation Type			Conf. Readout Pattern		Conf. Groups/Int		Conf. Integrations/Exp		Conf. Total Integrations		Conf. Total Exposure Time	
	1	c1			NRSIRS2RAPID		10		1		1		160.478	
	2	c2			NRSIRS2RAPID		10		1		1		160.478	

Proposal 7085 - Observation 3 - Dust Evolution in Molecular Cores

	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
Spectral Elements	1	1 (G140H/F100LP)	c1	3 Shutter Slitlet	295.281342125 Degrees 10.952416666666 666 Degrees	0.0013570470573 638764			3	3	2888.6
	2	2 (G235H/F170LP)	c1	3 Shutter Slitlet	295.281342125 Degrees 10.952416666666 666 Degrees	0.0013570470573 638764			3	3	2888.6
	3	3 (G395H/F290LP)	c1	3 Shutter Slitlet	295.281342125 Degrees 10.952416666666 666 Degrees	0.0013570470573 638764			3	3	2888.6
	4	1 (G140H/F100LP)	c2	3 Shutter Slitlet	295.27624929166 666 Degrees 10.947416666666 667 Degrees	3.8742159198750 246E-4			3	3	2888.6
	5	2 (G235H/F170LP)	c2	3 Shutter Slitlet	295.27624929166 666 Degrees 10.947416666666 667 Degrees	3.8742159198750 246E-4			3	3	2888.6
	6	3 (G395H/F290LP)	c2	3 Shutter Slitlet	295.27624929166 666 Degrees 10.947416666666 667 Degrees	3.8742159198750 246E-4			3	3	2888.6
Special Requirements	MSA Planned Aperture PA 0.0000 to 0.0000 Degrees (V3 221.4254303 to 221.4254303)										